Lane-Emden equation

1. Introduction

The theoretical study of stellar structure seeks to characterize the internal state of a star in a steady-state equilibrium. Such a configuration is fundamentally defined by three parameters: its total mass (M), its radius (R), and its luminosity (L). Luminosity is formally defined as the total radiant energy emitted by the star per second.

The structural objective needs the determination of the spatial distribution of the state variables, specifically density (ρ), pressure (P), and temperature (T), throughout the configuration, often referred to as the çomplete march.ºf physical variables. Here, we adopt the standard restrictions established in Chapter I and II of the book **An Introduction to the Study of Stellar Structure** by S. Chandrasekhar (1939), where radiation pressure is neglected in the equation of hydrostatic equilibrium, and the chemical composition is assumed to be dominated by hydrogen and helium. These simplifications allow for the derivation of a robust mathematical theory of the stellar interior.

2. Physical Principles of Hydrostatic Equilibrium

The mechanical equilibrium of a self-gravitating gas sphere is maintained by the balance between the inward pull of gravity and the outward gradient of internal pressure.

The Fundamental Equation: The equation of hydrostatic equilibrium for a spherically symmetric mass is expressed as:

$$\frac{dP}{dr} = -\frac{GM(r)}{r^2}\rho \tag{1}$$

where r is the radius vector from the center, G is the gravitational constant, and $M(r)$ is the mass inclosed within a spherical surface of radius r.

Mass Distribution: The relationship between the inclosed mass and the density distribution is governed by the mass conservation equation:

$$\frac{dM(r)}{dr} = 4\pi r^2 \rho \tag{2}$$

The Ideal Gas Law To link the mechanical variables to the thermal state, we assume the stellar material behaves as an ideal gas:

$$P = \frac{\mathcal{R}}{\mu}\rho T \tag{3}$$

In this context, $\mathcal{R}$ denotes the universal gas constant (distinct from the stellar radius R), and μ represents the mean molecular weight. This relation provides the equation of state necessary to close the system of equations.

3. Theory of Polytropic Configurations

Polytropic models simplify the complex relation between pressure and density by assuming a quasi-statical change of state.

The Polytropic Relation: Following Chandrasekhar, we define the polytropic exponent γ' in terms of specific heats:

$$\gamma' = \frac{c_p - c}{c_v - c} \tag{4}$$

where c is the specific heat maintained constant during the change. The polytropic index n is then defined as:

$$n = \frac{1}{\gamma' - 1} \tag{5}$$

This leads to the power-law relationship

$$P = K\rho^{1+\frac{1}{n}}, \tag{6}$$

where K is a constant of proportionality.

Pressure-Density Derivation: Using the "polytropic temperature" θ, where

$$T = T_c\theta \tag{7}$$

and T_c is the central temperature, and the relation

$$\rho = \lambda\theta^n, \tag{8}$$

the pressure-density link is established algebraically.
Since $P \propto \rho T$ and $T \propto \theta \propto \rho^{1/n}$:

$$P \propto \rho \cdot \rho^{1/n} = \rho^{1+\frac{1}{n}} \tag{9}$$

This polytropic assumption allows the structural equations to be solved without detailed knowledge of the energy generation mechanisms, provided the index n is specified.

4. Mathematical Derivation of the Lane-Emden Equation

To transform the physical equations of equilibrium into a dimensionless form, we introduce the Lane variables: the dimensionless temperature θ and the dimensionless radius ξ.

Dimensionless Variables and Scaling: We define the density in terms of the central density ρ_c:

$$\rho = \rho_c\theta^n \tag{10}$$

The physical radius r is scaled by a constant α:

$$r = \alpha\xi, \quad \text{where} \quad \alpha = \left[\frac{(n+1)K\rho_c^{1/n-1}}{4\pi G}\right]^{1/2} \tag{11}$$

4.1. Step-by-Step Substitution

1. **Differentiating the Equilibrium Equation:** Multiply the hydrostatic equation by r^2/ρ and differentiate with respect to r:

$$\frac{d}{dr}\left(\frac{r^2}{\rho}\frac{dP}{dr}\right) = -G\frac{dM(r)}{dr} \tag{12}$$

2. **Eliminating Mass:** Substitute the mass distribution equation

$$\frac{dM(r)}{dr} = 4\pi r^2\rho : \frac{1}{r^2}\frac{d}{dr}\left(\frac{r^2}{\rho}\frac{dP}{dr}\right) = -4\pi G\rho \tag{13}$$

3. **Applying Dimensionless Variables:** Substitute $P = K\rho_c^{1+1/n}\theta^{n+1}$ and $r = \alpha\xi$. The derivative $\frac{dP}{dr}$ becomes:

$$\frac{dP}{dr} = \frac{dP}{d\theta}\frac{d\theta}{d\xi}\frac{d\xi}{dr} = \left[(n+1)K\rho_c^{1+\frac{1}{n}}\theta^n\right]\frac{d\theta}{d\xi}\frac{1}{\alpha} \tag{14}$$

   Substituting these into the second-order form and simplifying using the definition of α yields the Lane-Emden equation.

The Lane-Emden Equation: The resulting dimensionless Poisson equation for a self-gravitating gas is:

$$\frac{1}{\xi^2}\frac{d}{d\xi}\left(\xi^2\frac{d\theta}{d\xi}\right) = -\theta^n \tag{15}$$

Subject to the central boundary conditions:

$$\theta(0) = 1 \text{ and } \left.\frac{d\theta}{d\xi}\right|_{\xi=0} = 0. \tag{16}$$

5. Analytical Solutions for Specific Indices ($n = 0,\ 1,\ 5$)

Solution for n = 0 (Uniform Density) When $n = 0$, the equation is

$$\frac{1}{\xi^2}\frac{d}{d\xi}(\xi^2\frac{d\theta}{d\xi}) = -1. \tag{17}$$

Integrating once:

$$\xi^2\frac{d\theta}{d\xi} = -\frac{1}{3}\xi^3 + C. \tag{18}$$

Since $\theta'(0) = 0$, entonces $C = 0$. Integrating again:

$$\frac{d\theta}{d\xi} = -\frac{1}{3}\xi \implies \theta(\xi) = 1 - \frac{\xi^2}{6}. \tag{19}$$

Solution for n = 1 (Linear Relation): For n=1, the equation becomes a linear spherical Bessel equation of order zero.

$$\text{Result: } \theta(\xi) = \frac{\sin \xi}{\xi}. \tag{20}$$

Solution for n = 5 (The Schuster Sphere): For n=5, the equation admits a solution of the form:

$$\text{Result: } \theta(\xi) = \left(1 + \frac{\xi^2}{3}\right)^{-1/2}. \tag{21}$$

While this configuration possesses an infinite radius, the total mass remains finite, representing a highly condensed "point-source"model.

Comparison of Boundary Values (ξ_1 where $\theta = 0$)

Polytropic Index (n)	Boundary Value (ξ_1)	Physical Interpretation
n = 0	$\sqrt{6}$	Uniform density distribution
n = 1	π	Finite radius; density $\propto \sin \xi/\xi$
n = 5	∞	Infinite radius; finite total mass

6. Summary of Results and Physical Implications

The Lane-Emden equation provides a powerful framework for modeling stellar interiors using the polytropic index as a single governing parameter.

Relationship of Polytropic Variables

Index (n)	Pressure Exponent $(1 + 1/n)$	Physical State
0	∞	Incompressible fluid
1	2.0	Simple linear density gradient
1.5	1.67	Monatomic gas (adiabatic)
3	1.33	Relativistic gas / Eddington model
5	1.2	Schuster sphere (infinite extent)

In conclusion, the development of the Lane-Emden equation represents the flowering of a complete mathematical theory from a physical problem. By determining the spatial distribution of state variables through dimensionless variables, theorists can approximate the internal structure of stars across various evolutionary stages without requiring detailed knowledge of energy generation, provided the polytropic index accurately reflects the underlying thermodynamics.